

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and debugged **C++ firmware** for **ESP32 microcontrollers**, implementing communication stacks for **Bluetooth Low Energy (BLE)** and **Wi-Fi**.
- Integrated **Modbus RTU over RS-485** drivers into embedded applications, enabling reliable communication with industrial sensors and energy meters.
- Authored **Python scripts** for automating firmware testing and device provisioning workflows on **Linux** systems, reducing manual setup time by approximately **30%**.
- Managed embedded software lifecycle from architecture to production deployment, including writing **unit and integration tests** for firmware modules.
- Implemented **TLS authentication** and secure coding practices for IoT devices, mitigating risks associated with public-key cryptography in constrained environments.
- Collaborated with hardware engineers and cloud developers to ensure seamless device-level logic integration with **AWS IoT Core** backend infrastructure.

SKILLS

Embedded Systems – C++, Embedded C, ESP32, ESP-IDF, Microcontrollers, RTOS, FreeRTOS, Bare-metal

Wireless Protocols – Bluetooth Low Energy (BLE), Wi-Fi, TCP/IP

Wired Protocols – Modbus RTU, RS-485, I2C, UART

Cloud IoT – AWS IoT Core, MQTT, REST APIs, WebSockets, Protobuf, Device Provisioning, OTA Updates

Tools & Development – Linux, Python, Git, Jenkins, Docker, Cross-compilation, Shell Scripting, CI/CD

Security – Public-Key Cryptography, TLS, Secure Boot, Secure Coding Practices

PROJECTS

Smart Home Device Connectivity Firmware (HCL Technologies)

- Designed and implemented a robust **BLE communication stack** for a smart home device, managing advertisement packets and connection events on an **ESP32-S3 microcontroller**.
- Developed **Python test automation scripts** to validate Wi-Fi provisioning sequences and ensure successful integration with cloud IoT platforms, covering over **50 test cases**.
- Wrote comprehensive **technical documentation** for firmware modules, including register maps for integrated sensors and API usage guidelines for subsequent cloud integration.

Industrial Sensor Modbus Integration (HCL Technologies)

- Engineered a **Modbus TCP driver** in **Embedded C++** to interface with an industrial energy meter, enabling real-time data acquisition from critical operational systems.
- Implemented error handling and retry mechanisms for **Modbus RTU** communication over **RS-485**, improving data reliability by an estimated **15%** in noisy environments.
- Configured and managed **cross-compilation toolchains** within a **Linux** environment for building firmware targeted at diverse embedded platforms.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)